

Lecture Scribe
CSE 400 – Fundamentals of Probability in
Computing
Tutorial 2 Solutions
School of Engineering and Applied Science
(SEAS), Ahmedabad University

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This lecture scribe reconstructs the reasoning and derivations used in solving the problems presented in Tutorial 2. Each problem is developed step-by-step so that the logical progression of the arguments can be followed during exam revision.

1 Q1. Random Point on a Line Segment

1.1 Problem Statement

A point is chosen randomly on a line segment of length L . The line is divided into two segments by this point. We must determine the probability that the ratio of the shorter segment to the longer segment is less than $1/4$.

1.2 Step 1: Mathematical Representation

Let x be the distance of the chosen point from one endpoint of the line.

Since the point lies on the line segment,

$$0 < x < L$$

The two resulting segments are:

First segment: x

Second segment: $L - x$

1.3 Step 2: Ratio of Segments

The ratio of the shorter segment to the longer segment is

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We want to find the probability that

$$R < \frac{1}{4}$$

1.4 Step 3: Case Analysis

The nature of the shorter segment depends on whether the point lies to the left or right of the midpoint $L/2$.

1.4.1 Case 1: $x \leq \frac{L}{2}$

In this case:

shorter segment = x

longer segment = $L - x$

Thus the condition becomes

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiplying both sides by $4(L-x)$:

$$4x < L - x \quad 5x < L \quad x < \frac{L}{5}$$

Therefore, the valid values in this case are

$$0 < x < \frac{L}{5}$$

1.4.2 Case 2: $x > \frac{L}{2}$

Now:

shorter segment = $L - x$

longer segment = x

Thus

$$\frac{L-x}{x} < \frac{1}{4}$$

Multiplying both sides by $4x$:

$$4L - 4x < x \quad 5x > 4L \quad x > \frac{4L}{5}$$

Thus the valid interval becomes

$$\frac{4L}{5} < x < L$$

1.5 Step 4: Valid Region

Combining both cases:

$$(0, L/5) \cup (4L/5, L)$$

Length of valid region:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

1.6 Step 5: Probability Calculation

Because the point is uniformly distributed along the segment,

$$P = \frac{\text{valid region}}{\text{total region}} \quad P = \frac{2L/5}{L} \quad P = \frac{2}{5}$$

2 Q2. Bayesian Decision Problem with Normal Distributions

Given

Two regions exist in an image:

White Region

$$X|W \sim N(4, 4)$$

Thus

$$\sigma_1 = 2$$

Black Region

$$X|B \sim N(6, 9)$$

Thus

$$\sigma_2 = 3$$

Prior Probabilities

$$P(B) = \alpha \quad P(W) = 1 - \alpha$$

A measurement is observed:

$$X = 5$$

The problem asks for the value of α such that the probability of classification error is the same regardless of the decision made.

2.1 Step 1: Equal Posterior Probabilities

Equal classification error implies

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

2.2 Step 2: Apply Bayes' Theorem

$$P(B|X = 5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha)f(5|W)}$$

Setting this equal to $1/2$ gives the equation needed to solve for α .

2.3 Step 3: Normal Density Evaluation

The normal probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Substituting parameters:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18} \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

2.4 Step 4: Substitution

Substituting into Bayes' expression:

$$\frac{\alpha e^{-1/18}}{\alpha e^{-1/18} + (1-\alpha)e^{-1/8} \frac{3}{2}} = \frac{1}{2}$$

Simplifying yields

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

2.5 Step 5: Solve for α

Using the approximation

$$e^{-5/72} \approx 0.93 \quad 6\alpha = 3\alpha + 2(1 - \alpha)(0.93) \quad 3\alpha = 1.86 - 1.86\alpha \quad 4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Thus

$$\alpha \approx 0.38$$

3 Q3. Optimal Placement of a Fire Station

3.1 Part (a): Finite Road

Let

$$X \sim \text{Uniform}(0, A)$$

with density

$$f_X(x) = \frac{1}{A}$$

The objective is to minimize

$$E[|X - a|]$$

3.1.1 Step 1: Express the Expectation

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

3.1.2 Step 2: Differentiate

The resulting expression simplifies to

$$\frac{2a^2 - 2aA + A^2}{2A}$$

Differentiating with respect to a :

$$\frac{1}{2A}(4a - 2A)$$

3.1.3 Step 3: Minimization

Set derivative to zero:

$$4a - 2A = 0 \quad a = \frac{A}{2}$$

Therefore, the station should be located at the midpoint of the road.

3.2 Part (b): Infinite Road

Now

$$X \sim \text{Exponential}(\lambda)$$

PDF:

$$f_X(x) = \lambda e^{-\lambda x}$$

The expected distance is minimized when

$$\frac{d}{da} E[|X - a|] = 1 - 2e^{-\lambda a}$$

Setting this equal to zero:

$$e^{-\lambda a} = \frac{1}{2}$$

Taking logarithms:
 $-\lambda a = \ln(1/2) \quad a = \frac{\ln 2}{\lambda}$

4 Q4. Mixture of Exponential Battery Lifetimes

Let X denote battery lifetime.

Two types exist with probabilities p_1 and p_2 .

Goal:

$$P(X > t + s | X > t)$$

Using total probability:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t} \quad P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Thus

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

5 Q5. Conditional Distribution of Poisson Variables

Let

$$X \sim \text{Poisson}(\lambda_1) \quad Y \sim \text{Poisson}(\lambda_2)$$

Independent.

We seek

$$P(X = k | X + Y = n)$$

Using conditional probability:

$$P(X = k | X + Y = n) = \frac{P(X=k, Y=n-k)}{P(X+Y=n)}$$

Substituting Poisson PMFs and simplifying results in

$$P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Thus the conditional distribution is binomial.

6 Remaining Problems (Conceptual Summary)

6.1 Q6

Transformation of a uniform random variable producing a distribution with:
 continuous density in certain intervals
 a discrete point mass at $Y = 0$ equal to $1/2$.

6.2 Q7

Stock price movements are modeled with independent Bernoulli trials, producing a Binomial distribution for the number of upward moves.

6.3 Q8

Repair time follows an Exponential distribution with parameter $\lambda = 1/2$.

Key results include:

$$P(X > 2) = e^{-1}$$

and

$$P(X \geq 10 | X > 9) = e^{-1/2}$$

6.4 Q9

For

$$Z = \sqrt{X^2 + Y^2}$$

the CDF is computed by integrating over a circular region $x^2 + y^2 \leq z^2$.

The PDF is obtained by differentiating the CDF using Leibniz's rule.

6.5 Q10

For the maximum of i.i.d. variables

$$M_n = \max(X_1, \dots, X_n)$$

the joint distribution of M_n and M_{n+1} depends on whether $x \leq y$ or $x > y$:

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$